

Hybrid Power Transformer Voltage Control Strategy for Grid-connected Photovoltaics

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Abstract—The access of new energy field stations to the distribution network is prone to voltage overruns, power fluctuations and other problems, affecting the safe and stable operation of the power system. Hybrid power transformers can achieve dynamic regulation of the target bus voltage through the regulation of power electronic devices to alleviate the power quality problems of the power system. In this paper, the equivalent circuit of hybrid power transformer is constructed, the mathematical model of hybrid power transformer voltage regulation is deduced, and after revealing the mechanism of PV grid-connected voltage overrun and voltage fluctuation, the hybrid power transformer voltage regulation control strategy for PV grid-connected scenarios is proposed. Simulation results show that the proposed control strategy can effectively alleviate the voltage overrun problem brought about by the instantaneous grid-connection of PV system, and the hybrid power transformer can control the target node voltage to be stable at the commanded value under the working conditions of PV output change and load surge.

Keywords—hybrid power transformer, photovoltaic grid connection, voltage regulation control, voltage overrun, voltage fluctuation

I. INTRODUCTION

With the wind power, photovoltaic and other new energy large-scale grid connection, the power system presents a significant trend of power electronics, and often appear near the grid point of high and low voltage overruns and other problems, leading to the emergence of new energy large-scale chain off-grid phenomenon, which seriously affects the safe and stable operation of the power system [1].

In order to suppress voltage overruns caused by grid connection or power fluctuations of new energy stations, hybrid power transformer (HPT) has been developed, which combines a traditional power transformer with a voltage source converter (VSC) that accounts for only 10-20% of the total capacity [2-3]. The traditional power transformer is used for power transmission, and the voltage source converter is used for voltage regulation, harmonic suppression, current regulation, and other regulation tasks [4-6]. HPT has the advantages of economy, reliability, and flexible regulation, which is of great

significance to improve the controllability and flexibility of the new type of power system.

When distributed PV is reasonably connected to the distribution network, it can improve the distribution network voltage distribution to a certain extent, but irrational access will change the system tidal current and lead to voltage quality deterioration. Different PV access locations, access capacity and power factor will have an impact on distribution network voltage quality. In order to solve the power system voltage fluctuation and voltage overrun problems, scholars at home and abroad have conducted relevant research on HPT voltage regulation control strategies. Reference [7] proposed a control strategy for hybrid transformers to dynamically regulate the grid voltage and improve the grid-side power factor, but the power convergence of this method is slow. Reference [8] proposed a hybrid distribution transformer with local and remote cooperative control that can solve the problems of voltage fluctuation and load distortion. Reference [9] proposed a decentralised model predictive control strategy for HPT voltage regulation with a robust tube-based compensation method that corrects abnormal voltages and reduces reactive power injection/absorption. The HPT control strategy proposed in the reference [10] can improve power quality problems such as voltage dips/spikes, current/voltage distortion, and current reversal, and it can prolong the life of the main transformer. Comprehensive analysis of the above, the current research on HPT voltage regulation and control strategy is mostly from the perspective of HPT performance, not combined with the voltage fluctuation mechanism of PV grid-connected, and the research on HPT to manage the power quality problems caused by new energy sources such as PV and other new energy sources connected to the power system is not yet clear.

Based on the HPT topology, this paper proposes to construct the HPT equivalent circuit that can effectively reflect the external output characteristics of the converter, derive the HPT voltage regulation mathematical model, analyse the voltage overrun and voltage fluctuation mechanism of the photovoltaic system access to the distribution network, and study the HPT voltage regulation and control strategy for the photovoltaic grid-connected voltage problem. To mitigate the voltage and power fluctuation problems caused by new energy grid-connection, we provide a means to solve the problem.

II. HPT EQUIVALENT CIRCUIT

The HPT consists of three parts: the main transformer, the shunt side converter VSC_sh and the series side converter VSC_se. the HPT main circuit topology is shown in Fig. 1.

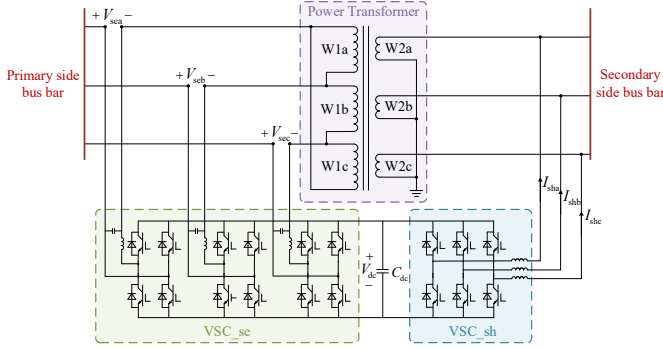


Fig. 1. Topology of HPT main circuit

In Fig. 1, the main transformer is a three-phase, two-winding conventional power transformer; VSC_sh is a three-phase bridge voltage source converter connected in parallel to the secondary side of the main transformer, which is used to maintain the stability of the DC voltage V_{dc} , reactive power compensation and harmonic current suppression. VSC_se are three single-phase full-bridge voltage source converters connected in series on the primary side of the main transformer for voltage regulation and reactive power compensation. VSC_sh and VSC_se are connected via DC capacitor C_{dc} and complete the active power exchange.

Neglecting the power electronic switching losses, on the one hand, VSC_sh can be equated to an ideal controlled current source since VSC_sh controls the current I_{sh} injected into the system. On the other hand, VSC_se controls the output voltage V_{se} , so that VSC_se can be equated to an ideal controlled voltage source. The resulting HPT equivalent circuit diagram is shown in Fig. 2.

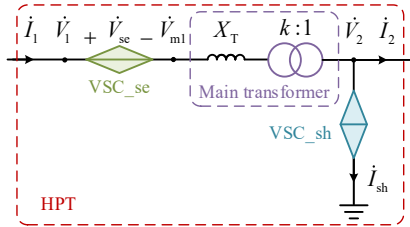


Fig. 2. Equivalent circuit of HPT

In Fig. 2, V_1 is the primary side voltage of the HPT, V_{m1} is the primary side winding voltage of the main transformer; V_2 is the HPT secondary voltage; I_1 is the current on the primary side of the HPT; I_2 is the current on the secondary side of the HPT; X_T is the equivalent leakage reactance of the main transformer; and $k:1$ represents the ratio of the main transformer's primary and secondary side winding voltages.

The expression of the relationship between the HPT winding current and winding voltage can be obtained from Fig. 2 as

$$\begin{cases} \dot{V}_1 = \dot{V}_{se} + k\dot{V}_1 + jX_T\dot{V}_1 \\ k\dot{I}_1 = \dot{I}_2 + \dot{I}_{sh} \end{cases} \quad (1)$$

If the HPT primary side current I_1 is used as the reference phase quantity. Setting the HPT primary side voltage as $\dot{V}_1 = V_1 \angle \theta_1$, the HPT secondary side voltage as $\dot{V}_2 = V_2 \angle \theta_2$, the output voltage of VSC_se as $\dot{V}_{se} = V_{se} \angle \theta_{se}$, and the output current of VSC_sh as $\dot{I}_{sh} = I_{sh} \angle \theta_{sh}$. According to Fig. 2 and (1), we can get the HPT phase relation diagram as shown in Fig. 3.

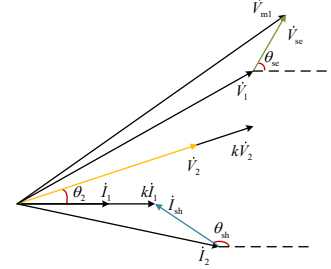


Fig. 3. Phase relationship diagram of HPT

Let the active power absorbed from the system by VSC_sh be P_{sh} and the active power injected into the system by VSC_se be P_{se} , so that there are

$$\begin{cases} P_{sh} = V_2 I_{sh} \cos(\theta_2 - \theta_{sh}) \\ P_{se} = V_{se} I_1 \cos \theta_{se} \end{cases} \quad (2)$$

Assuming that the HPT DC side losses are ignored, the active power absorbed by VSC_sh needs to have the following power balance with the active power injected into the system by VSC_se

$$V_{se} I_1 \cos \theta_{se} + V_2 I_{sh} \cos(\theta_2 - \theta_{sh}) = 0 \quad (3)$$

III. ANALYSIS OF PHOTOVOLTAIC GRID-CONNECTED VOLTAGE FLUCTUATION PROBLEM

As more and more distributed PVs are connected to the grid, the end distribution network in remote areas is prone to the phenomenon of high penetration of PVs, which generates voltage overruns and voltage fluctuations. The introduction of PV systems will reduce the voltage loss on the feeder and raise the voltage at each load node. The voltage fluctuation problem of distribution network containing PV system is analysed below. The schematic diagram of the distribution network containing PV system is shown in Fig. 4.

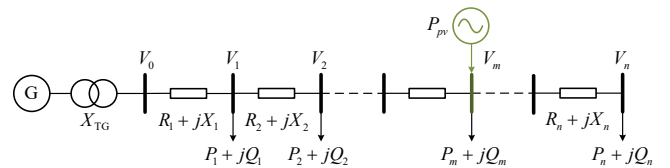


Fig. 4. Schematic diagram of the distribution network with photovoltaic power generation system

In Fig. 4, the distribution line contains a total of n node; V_0 , represents the initial voltage of the distribution feeder; the voltage of the i th node is $V_i (i=1,2,\dots,n)$; the line impedance is

$R_i + jX_i (i=1,2,\dots,n)$; with the direction of power flow from the bus to the load as the positive direction of power, the apparent power of the i th node is $P_i + jQ_i (i=1,2,\dots,n)$; and the PV system is set to be connected to the node m , and the capacity of the connected PV power is P_{pv} .

When the PV system is not connected to the distribution grid, if the losses on the line are neglected, the voltage drops at the k th and $(k-1)$ th nodes can be obtained as ΔV_{pv} .

$$\Delta V_k = V_{k-1} - V_k = \frac{\sum_{i=k}^n P_i R_k + \sum_{i=k}^n Q_i X_k}{V_{k-1}} \quad (4)$$

When the PV system is connected to the distribution network, if the PV system is connected after node $k (0 < k < m)$, the voltage at node k can be obtained as

$$V_{k1} = V_0 - \sum_{t=1}^k \frac{\left(\sum_{i=t}^n P_i - P_{pv} \right) R_t + \sum_{i=t}^n Q_i X_t}{V_{t-1}} > V_k \quad (5)$$

From (5), it can be seen that the node voltage increases after the distribution network is connected to the PV system. At this time, the voltage difference between node k and node m is

$$\Delta V_{k1} = V_{k-1} - V_k = \frac{\left(\sum_{i=k}^n P_i - P_{pv} \right) R_k + \sum_{i=k}^n Q_i X_k}{V_{k-1}} \quad (6)$$

From (6), it can be seen that distributed PV access to the distribution network will increase the node voltage, and the larger the PV grid-connected capacity is, the greater the impact on the node voltage. When the PV grid-connected capacity is too large, the voltage difference between neighbouring nodes may become negative.

If the PV system is connected before node $k (k > m)$, at this point, the voltage difference between node k and node $k-1$ is

$$\Delta V_{k2} = \sum_{t=m+1}^{k-1} \frac{\sum_{i=t}^n P_i R_t + \sum_{i=t}^n Q_i X_t}{V_{k-1}} - \sum_{t=m+1}^k \frac{\sum_{i=t}^n P_i R_t + \sum_{i=t}^n Q_i X_t}{V_{k-1}} \quad (7)$$

From (7), it can be seen that ΔV_{k2} is always greater than 0, and the node k -voltage is somewhat improved. However, if the node is further away from the distributed PV access point, the node voltage is lower. If the PV grid-connected capacity is large, it may cause voltage overrun problem near the grid-connected point.

The PV grid-connected point is usually the node most affected by the PV power supply, so node m is chosen as the object of study to analyse the impact of PV grid-connectedness on voltage fluctuations, and the Davignan equivalent circuit of the distribution network containing distributed PV is shown in Fig. 5.

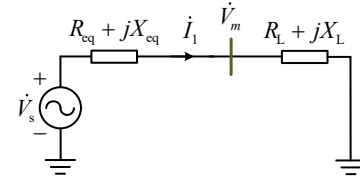


Fig. 5. Davignan Equivalent Circuit for Distribution Grids with Distributed Photovoltaics

In Fig. 5, $\dot{V}_s$ is the system power supply; $R_{eq} + jX_{eq}$ is the system equivalent impedance looking in from node m ; and $R_L + jX_L$ is the load-side equivalent impedance looking in from node m .

Assuming that the current variation due to power change after grid connection of the PV system is ΔI_L , the value of voltage fluctuation ΔV_m at node m is

$$\begin{cases} \Delta V_m = (R_{eq} + jX_{eq}) \Delta I_1 = \frac{V_m \Delta P_{pv}}{S_m} [\cos(\theta + \delta) + j \sin(\theta + \delta)] \\ Z_{eq} = R_{eq} + jX_{eq} = |Z_{eq}| \angle \theta \\ \Delta I_1 = |\Delta I_1| \angle \delta \end{cases} \quad (8)$$

where ΔP_{pv} is the amount of power fluctuation of the PV system; S_m is the node m capacity.

If the vertical component is neglected, the value of voltage fluctuation at the PV grid point is obtained as

$$\Delta V_m = \frac{V_m \Delta P_{pv}}{S_m} \cos(\theta + \delta) \quad (9)$$

Assuming that the voltage of the end power consumer is $\dot{V}_L$ and the power of the consumer load $S_L = P_L + jQ_L$, the load side current $\dot{I}_L$ is

$$\dot{I}_L = \frac{P_L - jQ_L}{\dot{V}_L} \quad (10)$$

From (10), if the load node to the head-end voltage needs to pass through r line nodes, then the head-end to load-end voltage landing ΔV_L is

$$\Delta V_L = \left(\sum_{i=1}^r R_i + j \sum_{i=1}^r X_i \right) \frac{P_L - jQ_L}{\dot{V}_L} \quad (11)$$

In summary, the PV system output power, power factor, the load capacity and system short-circuit capacity will affect the fluctuation of voltage at the PV grid-connected point. If the short-circuit capacity of the distribution network to which the PV system is connected is small, the PV grid connection will have a greater impact on the voltage fluctuation.

IV. VOLTAGE REGULATION CONTROL STRATEGY BASED ON HPT

In order to solve the voltage overrun problem caused by PV system access to the distribution network, and to alleviate the voltage fluctuation caused by the instantaneous grid connection of PV system, this paper adopts the voltage management scheme

of HPT access to the distribution network, which achieves the regulation of the distribution network node voltage by controlling the amplitude and phase angle of the output voltage $\dot{V}_{se}$ of the HPT. The equivalent circuit of the distribution grid containing distributed PV connected to the HPT is shown in Fig. 6.

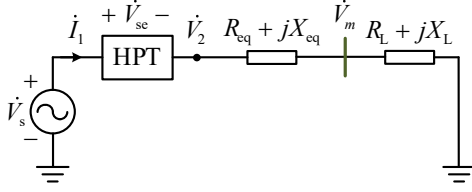


Fig. 6. Equivalent circuit of distribution grid with distributed PV connected to HPT

From Fig. 6, the expression for the voltage at node m after accessing the HPT can be obtained as

$$\dot{V}_m = \dot{V}_s - (R_{eq} + jX_{eq})\dot{I}_1 - \dot{V}_{se} \quad (12)$$

When the HPT carries out voltage regulation, the HPT its voltage regulation equation that is shown in (12), by changing the amplitude and phase angle of the HPT output voltage $\dot{V}_{se}$, the controlled bus voltage can be controlled at the commanded value to prevent the voltage from overrunning the limit.

From (1) and (3), the expressions for the HPT voltage and current relationships and the VSC_sh and VSC_se power balance relationships in the dq-synchronous rotating coordinate system are obtained as

$$\begin{cases} V_{1d} = V_{sed} + kV_{2d} \\ V_{1q} = kV_{2q} + X_T I_{1d} \\ kI_{1d} = I_{2d} + I_{shd} \\ kI_{1q} = I_{2q} \\ \frac{1}{2}V_{sed}I_{1d} + \frac{1}{2}(V_{2d}I_{shd} + V_{2q}I_{shq}) = 0 \end{cases} \quad (13)$$

where the physical quantity subscript d and subscript q denote the d-axis component and q-axis component, respectively, of the quantity in the dq-synchronous rotating coordinate system.

Without additional reactive output, $\dot{V}_2$ and $\dot{I}_{sh}$ are in the same phase. Assuming g be the intermediate variable, the linear expression for V_{sed} and g can be obtained as

$$g = -\frac{k^2 I_{1d} V_{sed}}{V_{1d}^2 - (V_{1q} - X_T I_{1d})^2} \quad (14)$$

Substitute (14) for (13), the expression for the relationship between V_2 and V_{sed} can be obtained

$$\begin{cases} V_{2d} = \frac{1}{k}(V_{1d} - V_{sed}) \\ V_{2q} = \frac{V_{1q}}{k} - \frac{X_T}{k^2} \left[I_{2d} - \frac{kI_{1d}V_{1d}V_{sed}}{V_{1d}^2 - (V_{1q} - X_T I_{1d})^2} \right] \end{cases} \quad (15)$$

From (15), both V_{2d} and V_{2q} change with V_{sed} . Therefore, by controlling the amplitude and phase angle of $\dot{V}_{se}$, the regulation of the system node voltage can be achieved.

Let V_{2_rms} be the rms value of the HPT secondary side voltage, $V_{2_rms_ref}$ be the command value of the voltage control target, and the steady-state value of the d-axis component V_{sed} of the VSC_se output voltage is expressed as

$$V_{sed} = \frac{V_{2_rms_ref}^2 - V_{2_rms}^2}{V_{1d}} + \frac{V_{1_rms}^2 - V_{2_rms}^2}{V_{1d}} \quad (16)$$

By adding the PI controller in (16), the HPT voltage regulation control strategy is obtained as

$$V_{sed} = \frac{\left(k_{p1} + \frac{k_{i1}}{s}\right)(V_{2_rms_ref}^2 - V_{2_rms}^2)}{V_{1d}} + \frac{V_{1_rms}^2 - V_{2_rms}^2}{V_{1d}} \quad (17)$$

where k_{p1} and k_{i1} are the proportional and integral coefficients of the PI control link in the HPT voltage regulation control strategy, respectively.

According to the proposed HPT voltage regulation control strategy, after determining the command value of the HPT voltage regulation control, V_{sed} can be calculated so that VSC_se inverts the corresponding output voltage through SPWM modulation to realise the function of the HPT to regulate the bus voltage.

V. SIMULATION ANALYSIS

In order to verify the effectiveness of the proposed HPT voltage regulation control strategy, a distribution network simulation model with HPT and PV system is constructed in PSCAD/EMTDC as shown in Fig. 7, which contains the system supply voltage, HPT, line, PV system and load. The simulation analysis is carried out through two working conditions: (1) light intensity and temperature change and (2) load surge.

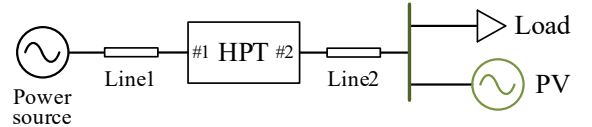


Fig. 7. Distribution network simulation model with HPT and PV systems

The main parameters of the simulation model system are shown in Table 1.

TABLE I. Main Parameters of the Simulation Modelling System

Parameters	Value
System supply voltage	10.5kV
HPT Rated Ratio	10kV/0.4kV
HPT Rated Capacity	1.6MW
HPT leakage resistance	0.04pu
HPT secondary side voltage command value	1.05pu
DC Capacitor Voltage	1.0kV
DC capacitance	1000 μ F
Line 1 impedance	0.120+j0.534 Ω
Line 2 impedance	0.02+j0.041 Ω
Initial output power of PV	0.384MW
Constant power load	0.3MW

1) Light intensity and temperature change conditions

In order to verify the effectiveness of the proposed HPT voltage regulation control strategy under the changing conditions of light intensity and temperature, the following simulation process is set up: the total simulation time is 2.0s. 0.3s the HPT is put into operation, 0.5s the main circuit of the PV system is put into operation, and 0.7s the control part of the PV system is put into operation. 1.0s the light intensity is increased from 1000W/m^2 to 2000W/m^2 and the temperature is increased from 25°C to 35°C . 1.5s the light intensity is decreased from 2000W/m^2 to 500W/m^2 .

The HPT secondary side voltage waveform is shown in Fig. 8. Under HPT voltage control, the secondary side voltage fluctuates briefly at the moment of PV input. 1.0s and 1.5s moments due to changes in light intensity and temperature, the secondary side voltage fluctuates and then restores the command value of 1.05 pu to run stably, and the fluctuation duration is within 0.1s. However, if there is no HPT control, then the secondary side voltage in the photovoltaic system into the moment of a large voltage drop. And the voltage always changes with the change of PV output, and can not be stabilised within the standard range. The proposed control strategy reduces the response time of voltage regulation by about 6%.

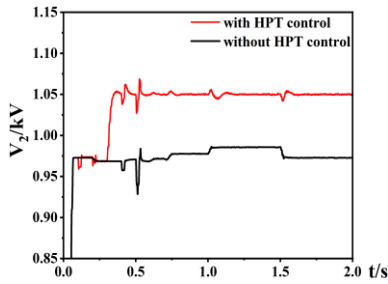


Fig. 8. HPT secondary voltage under varying light intensity and temperature conditions

The waveforms of active power, reactive power, DC voltage and output voltage at the secondary side of the HPT are shown in Fig. 9.

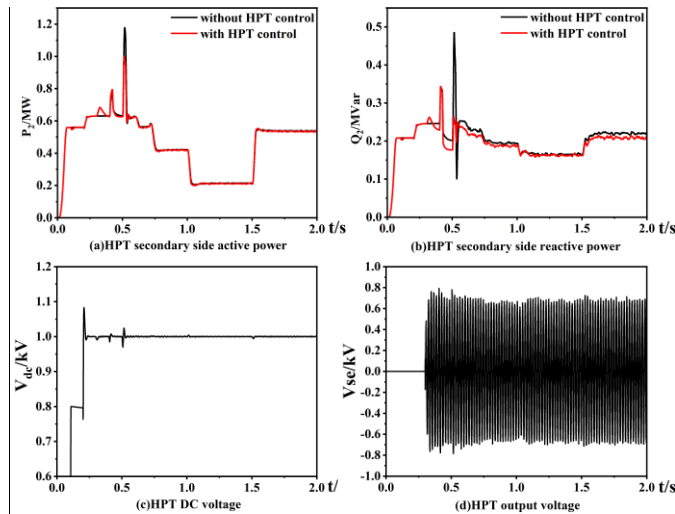


Fig. 9. The waveforms of active power, reactive power, DC voltage and output voltage at the secondary side of the HPT

In Fig. 9(a) and 9(b), at the instant of PV grid-connection at the moment of 0.5s, the active power and reactive power rise to 1.177MW and 0.485MVar, respectively, and a large shock occurs. If the voltage regulation control is applied, the active and reactive power only rises to 1.001 MW and 0.343 MVar, and the shock is effectively mitigated. In Fig. 9(c), the HPT completes uncontrolled rectifier charging and active charging before 0.3 s. A 2% overshoot occurs at the moment of 0.5 s for the PV grid connection, and a very small fluctuation occurs instantaneously for the change of PV output, and the rest of the time maintains the commanded value of 1.0 kV for stable operation. In Fig. 9(d), the HPT is used to stabilise the target bus voltage at 1.05 pu during 1.0-1.5 s, so the output voltage is 0.695 kV. After 1.5 s the PV output changes again and the output voltage becomes 0.943 kV.

Comprehensive analysis of the above, HPT can achieve rapid and continuous regulation of voltage, to make up for the shortcomings of traditional power transformer tap step regulation. At the moment of PV grid-connection, HPT can effectively alleviate the voltage and power impact brought about by the grid-connection instant. When the PV output occurs, the HPT makes the controlled node voltage always stable at the command value of 1.05 pu. The above verifies the effectiveness of the proposed HPT voltage regulation in the control strategy under the PV output change condition.

2) Load surge conditions

The following simulation flow is set up under the load surge condition: the total simulation time is 2.0s. 0.3s HPT is put into operation. 0.3MW active load and 0.06MVar reactive load are put into the system at the moment of 0.8s, and 0.3MW load is put into the system at the moment of 1.4s.

The HPT secondary side voltage waveform is shown in Fig. 10. Under the HPT voltage control, 0.3-0.4s the HPT secondary side voltage rises from 0.969pu to the commanded value of 1.05pu. the commanded value is kept in operation during the period of 0.4-0.8s. at the moment of 0.8s, a short fluctuation of 0.06s occurs due to the load inputs and then it is kept in stable operation. at the moment of 1.4s, due to the large load of the system, the fluctuation of the secondary side voltage lasts for 0.18s and then it stabilizes at the command value. The proposed control strategy reduces the response time of voltage regulation by about 10% compared with no HPT control.

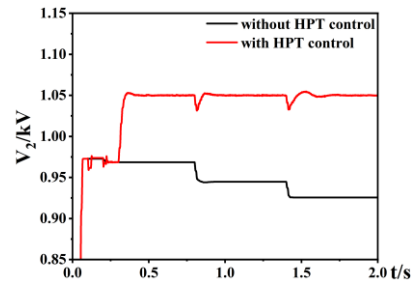


Fig. 10. HPT secondary side bus voltage under load surge condition

The waveforms of HPT secondary side active power, reactive power, DC voltage and output voltage are shown in Fig. 11.

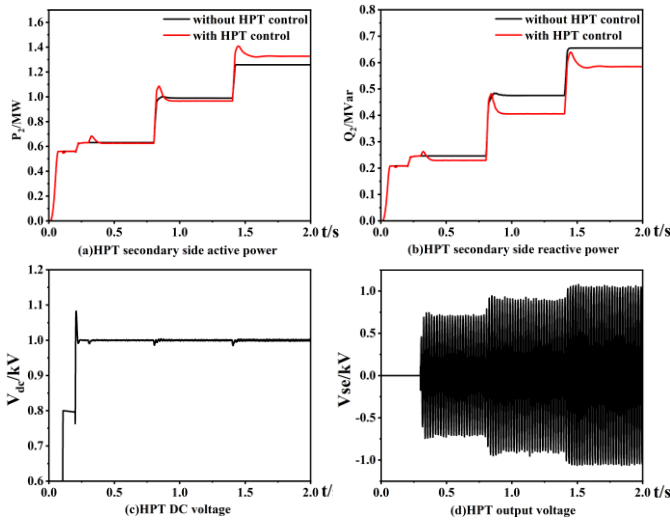


Fig. 11. The waveforms of active power, reactive power, DC voltage and output voltage at the secondary side of the HPT under load surge conditions

In Figures 11(a) and 11(b), the active and reactive power on the secondary side of the HPT increases at the moment of load input. When the HPT applies voltage regulation control, the power fluctuates briefly at the moment of load change, and the active power rises to 0.966MW and the reactive power rises to 0.406MVar at the moment of 0.8s. At the moment of 1.4s, the active power rises to 1.327MW, and the reactive power rises to 0.584MVar. When there is no voltage regulation control of the HPT, the active power rises to 0.990MW and the reactive power rises to 0.990MW at the moment of 0.8s. 0.990MW and reactive power rises to 1.258MW and reactive power rises to 0.655MVar. In Fig. 11(c), the 0.8s and 1.4s moments have small fluctuations due to load input, and the rest of the time maintains the commanded value of 1.0kV for stable operation. In Fig. 11(d), the HPT output voltage remains zero until 0.3 s. After 0.3 s, the output voltage rises with the voltage regulation and stabilises at 0.745 kV. At the moment of 0.8 s, the output voltage rises due to a sudden change in the load and stabilises at 0.947 kV. At the moment of 1.4 s, due to the input of yet another load, the output voltage rises again and stabilises at 1.069 kV.

After the above analyses, the HPT can quickly adjust the VSC_{se} output voltage to restore the bus voltage to the set value after being disturbed by the load input. This verifies the effectiveness of the proposed HPT voltage regulation control strategy under load surge conditions.

VI. CONCLUSION

In view of the voltage overrun and voltage fluctuation caused by new energy sources such as photovoltaic in the distribution network, this paper firstly constructs the corresponding equivalent circuit based on the HPT topology, researches the mathematical model of HPT voltage regulation, reveals the voltage overrun and voltage fluctuation mechanism of the photovoltaic system accessing the distribution network, and puts

forward the HPT voltage regulation and control strategy for the photovoltaic grid-connected voltage, and obtains the following conclusions by the simulation validation. The following conclusions are obtained through simulation verification:

(1) PV system output power, power factor, system short circuit capacity and distribution network equivalent impedance are all factors affecting the voltage fluctuation of PV grid-connected points. When distributed PV is connected to the distribution network, due to the relatively small short-circuit capacity of the distribution network, the PV grid connection will have a greater impact on the voltage fluctuation;

(2) The proposed HPT voltage regulation control strategy can achieve rapid and continuous regulation of voltage, making up for the shortcomings of traditional power transformer tap-type step voltage regulation; the proposed control strategy can effectively alleviate the voltage drop brought by the PV system grid-connected instantly to the power system; in the case of PV power change and load surge conditions, the HPT can make the voltage of the controlled node always stable at the commanded value of 1.05 pu by controlling the output voltage of VSC_{se}. 1.05pu.

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